

CONFIDENCE-AWARE LOCAL NAVIGATION

A lightweight ROS 2 architecture for a GPS-independent autonomous-drone MVP

CORE IDEA

Instead of treating sensor measurements as equally trustworthy at all times, the ROS 2 navigation layer evaluates local state confidence and adapts the drone's behavior.

Navigate when confident. Slow when uncertain. Stop when unsafe.

Project positioning

This is an MVP for a ROS course project, not a production-ready or research-grade GPS-denied flight system. The scope focuses on sensing → state estimation → confidence assessment → navigation → obstacle safety → PX4.

1. RESEARCH BASIS

What related systems establish

Approach	Main idea	Implication for our MVP
Wang et al.	Vision + IMU fusion	Improves local estimation but adds perception complexity.
Youn et al.	IMU + RGB-D/LiDAR + ESKF + SLAM	Strong benchmark, but research-grade complexity is unnecessary for the course MVP.
Lin & Zhan	VIO + UWB constraints	Complementary measurements reduce drift, but UWB adds infrastructure.
Lu et al.	Integrated perception + planning + control	Supports an end-to-end modular architecture.
Perez-Grau et al.	Localization + planning + obstacle avoidance	Shows navigation and safety should be integrated.
PX4 architecture	Optical-flow/range estimation + ROS 2/uXRCE-DDS	Provides a practical bridge between sensing, ROS 2 and flight control.

Not novel individually: IMU + optical flow + range, EKF/state estimation, ROS 2, PX4, obstacle avoidance or GPS-denied navigation.

Defensible differentiation: a lightweight confidence-aware ROS 2 navigation architecture tailored to a resource-constrained educational MVP.

2. PROPOSED APPROACH

Confidence-Aware Local Navigation (CALN)

A lightweight decision layer sits between state estimation and navigation. It uses sensor validity, quality, timeout and consistency signals to select an operating mode.

NORMAL NAVIGATION	Confidence high → move toward goal at normal velocity.
CAUTIOUS NAVIGATION	Confidence degraded → reduce velocity and use a conservative safety response.
SAFE HOLD	Confidence too low or safety compromised → stop/hover and wait for a valid state.

Do not claim to invent confidence-aware sensor fusion. Recent work already explores adaptive confidence-weighted localization. The MVP contribution is applying a lightweight confidence-driven safety/navigation policy inside a compact ROS 2 system.

3. MVP SYSTEM ARCHITECTURE

Keep the implementation small: seven meaningful ROS 2 nodes

Input	IMU • optical-flow/local-motion aid • range sensor
sensor_node	Publishes /sensors/imu, /sensors/flow, /sensors/range
state_estimator	Publishes /state/odom and /state/confidence
confidence_manager	Maps confidence to NORMAL / CAUTIOUS / SAFE_HOLD via /navigation/mode
goal_manager	Receives /navigation/goal and maintains target
navigation_node	Uses local state + target + mode → /navigation/cmd
obstacle_monitor	Range → CLEAR / WARNING / BLOCKED / INVALID
px4_bridge	Applies safe commands and communicates with PX4 through ROS 2/uXRCE-DDS

Core ROS 2 flow

/sensors/* → state_estimator → /state/odom + /state/confidence → confidence_manager → /navigation/mode → navigation_node + obstacle_monitor → /navigation/cmd → px4_bridge → PX4

4. PROPOSED MVP METHODOLOGY

A closed-loop, confidence-aware navigation workflow

1	Sensor acquisition	Read IMU, local-motion and range measurements.
2	State estimation	Estimate local pose/velocity from onboard measurements.
3	Confidence assessment	Check validity, quality, timeout and consistency.
4	Goal navigation	Generate simple movement toward the destination.
5	Obstacle monitoring	Classify path as CLEAR, WARNING, BLOCKED or INVALID.
6	Adaptive command	Move normally, slow down, stop/hover, or hold safely.
7	PX4 control	Send safe motion/setpoint commands through the flight-control interface.

Operating policy

HIGH confidence + CLEAR → NORMAL

MEDIUM confidence or WARNING → CAUTIOUS

LOW confidence, BLOCKED or INVALID → SAFE HOLD

5. ADAPTIVE SAFETY LOGIC

Safety is graded, not binary

Condition	State	Behavior
Obstacle far / clear	CLEAR	Normal navigation
Obstacle approaching	WARNING	Reduce velocity / increase caution
Obstacle inside critical zone	BLOCKED	Stop / hover
Sensor invalid or stale	INVALID	Safe hold / do not continue blindly
Localization confidence low	SAFE_HOLD	Stop until state becomes valid

Core design principle

Navigate when confident. Slow when uncertain. Stop when unsafe.

6. NOVELTY POSITIONING & MVP SCOPE

How to present this without overclaiming

Recommended claim

A lightweight confidence-aware ROS 2 navigation architecture tailored for a resource-constrained educational UAV MVP.

Do not claim

- A novel GPS-denied navigation algorithm
- Novel EKF / optical-flow fusion
- Novel obstacle avoidance
- Novel ROS 2 or PX4 integration

Do claim

- Compact integration of established sensing and navigation components
- Explicit confidence-driven operating modes
- Graded obstacle and estimator safety behavior
- A demonstrable ROS 2 architecture designed around MVP constraints

Current MVP

ROS 2 nodes + sensor processing + local state + simple goal navigation + range-based obstacle response + safe states + PX4 interface where feasible.

Future expansion

Visual-inertial odometry → 3-D SLAM → UWB/LiDAR fusion → dynamic obstacle prediction → learning-based navigation → larger GPS-denied experiments.

7. KEY REFERENCES

Supporting literature and official technical documentation

- [1] Chang et al. (2023), *A review of UAV autonomous navigation in GPS-denied environments*, Robotics and Autonomous Systems, 170, 104533.
- [2] Gyagenda et al. (2022), *A review of GNSS-independent UAV navigation techniques*, Robotics and Autonomous Systems, 152, 104069.
- [3] Youn et al. (2021), *Collision-free Autonomous Navigation of a Small UAV Using Low-cost Sensors in GPS-denied Environments*, International Journal of Control, Automation and Systems, 19(2), 953–968.
- [4] Lu et al. (2022), *Flight in GPS-denied environment: Autonomous navigation system for micro-aerial vehicle*, Aerospace and Science Technology, 124, 107521.
- [5] Lin & Zhan (2022), *GNSS-denied UAV indoor navigation with UWB incorporated visual inertial odometry*, Measurement, 201, 112256.
- [6] Perez-Grau et al. (2018), *An architecture for robust UAV navigation in GPS-denied areas*, Journal of Field Robotics, 35, 121–145.
- [7] PX4 documentation: ROS 2 / uXRCE-DDS integration and estimator documentation.
- [8] Recent adaptive confidence-weighted UAV localization work (2025): supporting context for confidence-driven measurement handling, not evidence that the general concept is novel.

Bottom line: the strongest contribution for a ROS course is not a new research algorithm. It is a small, coherent, failure-aware ROS 2 system that can actually be built, demonstrated and evaluated.